EvoStructCLIP: Multimodal Fusion of Structural and Evolutionary Features for Predicting TSC2 Protein Stability

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I. Data sources

For supervised training, we used experimentally measured variant abundance data from MaveDB. Specifically, we included two large-scale deep mutational scanning (DMS) datasets generated using Variant Abundance by Massively Parallel Sequencing (VAMP-seq): PTEN (urn:mavedb:00000013-a-1) and TPMT (urn:mavedb:00000013-b-1). VAMP-seq quantifies the cellular abundance of protein variants using fluorescence-based sorting and next-generation sequencing, providing a proxy for protein stability and loss-of-function effects [1].

To improve generalization, we performed pretraining on a large pathogenicity dataset curated from ClinVar [2]. We downloaded all human variants and filtered for missense variants with unambiguous pathogenic or benign labels. Quality control included mapping to canonical UniProtKB isoforms, sequence consistency checks using Swiss-Prot and TrEMBL reference FASTA files [3], structure mapping to AlphaFold DB models [4], and removal of duplicates. This yielded 153,787 unique missense variants with pathogenicity labels.

Multiple sequence alignments (MSAs) of homologous sequences were generated using MMseqs2 [5], and evolutionary descriptors (e.g., conservation, entropy) were derived from these MSAs. Structural information was obtained from AlphaFold DB release v4 [4].

II. Feature extraction

A. Voxel-based structural features

We generated three-dimensional voxel representations centered on the Cα atom of the mutated residue. A cubic grid of size $7 \times 7 \times 7$ with a spacing of 2 Å was defined, and each voxel was annotated with multiple structural descriptors. First, 42 “closeness” channels captured the proximity of CA and CB atoms from each of the 21 standard amino acids, following the general strategy of PrimateAI-3D [6] but extended with additional features. Relative sequence position was encoded by normalized signed and absolute distances between the mutated residue and the residue mapped to each voxel. Per-residue AlphaFold confidence scores (pLDDT) were projected onto the grid to reflect model reliability. Finally, local dynamic flexibility was quantified by mean square fluctuations derived from Gaussian Network Model (GNM) analysis using ProDy [8]. Together, these features produced approximately 46 channels per voxel grid, offering a richer structural representation than the original voxelization approach.

B. Evolutionary features

Evolutionary constraints were captured using multiple sequence alignments (MSAs) of homologous proteins generated with MMseqs2 [5]. MSAs were filtered by sequence identity, coverage, and redundancy thresholds to ensure high-quality alignments. From these MSAs, we derived conservation scores, amino acid substitution preferences, and position-specific information content, which provided complementary evolutionary context to the structural representations.

C. Biophysical and structural descriptors

For stability-specific prediction tasks, we extracted a compact set of residue- and mutation-level descriptors directly from AlphaFold models and MSA-derived statistics. These included pLDDT confidence scores at the mutated residue and their spatial neighbors (mean, minimum, maximum), mean square fluctuations from Gaussian Network Model (GNM) analysis [8], and long-range contact order calculated as the average sequence separation between spatially proximal residues. Physicochemical property changes upon mutation were quantified, including differences in residue volume, hydropathy based on the Kyte–Doolittle scale [9], and electrostatic charge. We further incorporated energetics features such as predicted ΔΔG_fold values computed with EvoEF2 [10], backbone torsion angles (φ and ψ) represented as sine and cosine terms, and binary indicators for substitutions involving glycine or proline. In addition, we integrated evolutionary conservation features derived from multiple sequence alignments, specifically ΔPSIC scores [7] and Shannon entropy at the mutated position. Altogether, this yielded a 19-dimensional feature vector per variant, designed to complement the voxel-based and evolutionary representations.

III. Model architecture (EvoStructCLIP, pretraining)

A. Overall framework

We developed EvoStructCLIP, a multimodal architecture designed to integrate structural and evolutionary evidence for missense variant effect prediction. The model contains two branches: a voxel-based encoder that captures local three-dimensional structural context around the mutated residue, and an MSA-based encoder that leverages evolutionary variation across homologous sequences. These branches produce embeddings that are aligned through a CLIP-style contrastive objective, while a classification head predicts pathogenicity [13].

B. Voxel branch

The voxel encoder processes the 3D voxelized structural features described earlier. The backbone is constructed from 3D MBConv blocks, which combine pointwise expansion, depthwise convolution, and squeeze-and-excitation attention [19,20]. This design allows the network to capture both fine-grained local interactions and broader spatial contexts while maintaining computational efficiency, consistent with principles from EfficientNet [11]. After global average pooling, the output is enriched with mutation-specific embeddings: both the wild-type and mutant residues are embedded, concatenated, and passed through non-linear transformations. These embeddings are added to the pooled structural representation, ensuring that the final vector explicitly encodes both the local structural context and the residue substitution itself.

C. MSA branch with cross-axial Mamba

The MSA encoder models evolutionary information from sequence alignments centered on the mutated position. Each amino acid is first embedded into a continuous space, and the resulting tensor has shape (L, D, C), where L is the alignment length window, D the MSA depth, and C the embedding dimension. To capture dependencies, we introduced a cross-axial Mamba block. Along the sequence length (L-axis), a full Mamba state-space layer processes each column, enabling long-range context propagation that is more efficient than classical recurrent or attention-based encoders [12,22]. Along the alignment depth (D-axis), convolutional filters are applied only at the central sequence position corresponding to the mutated residue, extracting local consensus signals across homologs. By combining outputs from both axes, the encoder integrates positional conservation with depth-wise evolutionary diversity in a complementary fashion [23]. Residual connections and RMS normalization stabilize training [24], and the final embedding is obtained by averaging the depth dimension at the mutation site.

D. Fusion and contrastive alignment

The voxel and MSA embeddings are L2-normalized and compared in a similarity matrix within each batch. A temperature-scaled CLIP objective encourages embeddings from the same variant to align closely while mismatched pairs are pushed apart, enforcing cross-modal consistency [13]. For downstream classification, the embeddings are either concatenated or added, followed by a lightweight feed-forward network with batch normalization and non-linear activation, outputting a scalar prediction score.

E. Training strategy

The model was trained end-to-end with a composite loss consisting of binary cross-entropy for pathogenicity, the CLIP contrastive loss [13], and an auxiliary FuseMix loss. FuseMix interpolates pairs of voxel and MSA embeddings in latent space using a shared Beta-distributed coefficient across modalities, ensuring that the augmented samples remain semantically consistent multimodal pairs. This latent-space augmentation alleviates data scarcity and improves robustness [14,25]. Training employed the AdamW optimizer [26] with cosine annealing learning rate scheduling [27] over 100 epochs. Early stopping was guided by validation performance using precision–recall AUC [28]. Data augmentations included random voxel rotations and flips [29], as well as MSA depth shuffling to prevent overfitting to alignment ordering.

IV. Downstream stability prediction

To adapt EvoStructCLIP for quantitative stability prediction in the TSC2 dataset, we employed a meta-feature regression strategy. For each variant, we concatenated the sigmoid-activated pathogenicity score produced by the pretrained EvoStructCLIP model with the 19-dimensional handcrafted biophysical descriptors described earlier [21]. This yielded a unified 20-dimensional feature vector per variant.

We trained gradient boosted decision trees using XGBoost [15] to map these meta-features into continuous stability values, where 0 corresponds to unstable, 1 to wild-type-like stability, and values greater than 1 indicate variants predicted to be more stable than the wild type. Hyperparameters—including the number of estimators, learning rate, maximum depth, child weight, subsampling, and regularization terms—were optimized with randomized search and five-fold cross-validation, using the Pearson correlation coefficient between predicted and experimental stability scores as the selection criterion.

Following hyperparameter optimization, a single best-performing model was retrained on the full training data, and an ensemble of five cross-validation models was also trained for uncertainty estimation. Predictions from the CV ensemble were aggregated by computing both the mean and standard deviation across folds, enabling more reliable stability estimates in addition to point predictions.

This strategy leverages EvoStructCLIP as a representation learner while employing a flexible non-linear regressor to integrate multimodal features into quantitative stability predictions. The resulting framework was applied to the held-out test set for submission to the CAGI challenge.

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